

Integration by Substitution

Date _____ Period _____

Evaluate each indefinite integral. Use the provided substitution.

1) $\int \frac{20x^4}{4x^5 + 3} dx; \quad u = 4x^5 + 3$

2) $\int 36x^2 e^{4x^3 + 3} dx; \quad u = 4x^3 + 3$

3) $\int 80x^3 \cdot 3^{5x^4 - 2} dx; \quad u = 5x^4 - 2$

4) $\int \frac{2}{x(-1 + \ln 4x)} dx; \quad u = -1 + \ln 4x$

Evaluate each indefinite integral.

5) $\int \frac{12x^2}{x^3 + 2} dx$

6) $\int \frac{20e^{5x}}{e^{5x} + 3} dx$

7) $\int 10 \sin -2x \cdot e^{\cos -2x} dx$

8) $\int \frac{5e^{-3 + \ln 3x}}{x} dx$